

When Is Global EOF Reduction Insufficient in Multi-Fidelity Flood Inundation Emulation?

Working title (methods paper). Hydrodynamically zoned LSG-Max under equal mode budget, area-weighted metrics, and train-only zoning.

Status: English manuscript draft v0.4 (Track B evidence locked; ≥ 5 ChatGPT co-iteration rounds + nature-polishing; public code at github.com/Coucou2016/202606-JOH-zonal-LSG). Publication figures cited here are `fig01–fig04` and `fig08–fig19` under `outputs/figures/` (SciencePlots + Times New Roman for Track B curves; `fig02` kept from real geometry export). Intermediate filenames `fig05–fig07*` and other report-only panels are reserved for the Chinese full report / SI and are intentionally not cited in this English draft. **nature-writing / polishing axes:** `task=manuscript`, `paper_type=methods`, `language=en`, `journal=generic` (JOH primary / WRR methods paradigm). **One-sentence argument:** Spatial zoning is not universally advantageous, but a global EOF representation can be performance-sensitive under constrained retained-mode capacity; under an audited retained-mode-budget protocol, zoning value is conditional on how spatial structure enters the coupled reduced-representation and LF-to-HF mapping pipeline, and first-order EOI alone cannot decide when to zone.

Abstract

High-resolution two-dimensional flood models remain expensive for ensembles and real-time use. Multi-fidelity surrogates in the Low-fidelity–Spatial analysis–Gaussian Process (LSG) family upskill coarse hydrodynamic fields by Empirical Orthogonal Function (EOF) reduction and Gaussian Process (GP) coefficient mapping. Baseline LSG formulations commonly employ a global EOF representation over the wet floodplain, but the consequences of changing that representation under a matched retained-mode budget have not been systematically isolated in the LSG studies considered here. We test that consequence with hydrodynamically zoned LSG-Max on the public Fraehr et al. benchmark under an audited retained-mode-budget protocol (primary matched-budget inference at $B = 4$), area-weighted depth metrics, and train-only zoning/EOF/GP fits. At Carlisle with budget $B = 4$, rule zoning reduces area-weighted RMSE from 0.1464 m (global) to 0.0964 m, with improvement on 9/9 event leave-one-out folds. The official two-fold split is not significant and is reported as a benchmark-compatible sensitivity check, not as the primary claim. Burnett 30-fold leave-one-out does not favour rule zoning; at Chowilla, LSG degrades relative to the low-fidelity field alone (an upstream applicability boundary for LSG correction). Across the three cases, max-surface residual organization indices ($EOI = 0.057 / 0.116 / 0.957$) do not rank zoning gains; second-order diagnostics, together with a Carlisle EOF $\times$ GP stage-swap, reject a pure-EOF-truncation explanation and are consistent with zonal structure acting through the coupled

representation–mapping pipeline. The contribution is a conditional, mechanism-aware reading of when global EOF capacity is insufficient, not a universal zoning recipe.

Keywords: multi-fidelity surrogate; flood inundation; EOF; Gaussian Process; LSG; zonal reduction; equal-budget comparison

1. Introduction

Fine-grid two-dimensional hydrodynamic models remain the reference for flood extent and depth, yet their cost still limits large ensembles, scenario design, and operational refresh cycles (Teng et al., 2017; Bates, 2022). Multi-fidelity strategies retain a physically based low-fidelity (LF) run and learn a correction toward high-fidelity (HF) fields from a modest training set. The LSG family implements this idea by projecting LF and HF inundation onto EOF spatial modes and training sparse or standard GPs on expansion coefficients (Fraehr et al., 2022, 2023, 2024). Related work spans deep-learning inundation surrogates and hybrid LF–GP pipelines (Bentivoglio et al., 2022; Lu et al., 2025). Baseline LSG implementations have predominantly used a single EOF domain, although recent studies have begun to regionalize training or spatial representation.

That global basis is convenient; it is not necessarily a performance-neutral representation choice for depth emulation. Channel, frequently inundated shelves, and fringe shallow water can mix into shared leading modes. When the retained-mode budget B is tight, such mixing may allocate limited retained-mode capacity inefficiently and alter downstream depth learning even when wet–dry skill looks acceptable. Prior work already shows that spatial reduction, rotated/localized EOF structure, and LSG regionalization can help in focused settings: PCA–ANN downscaling (Carreau and Guinot, 2021), SRR/USRR reconstruction (Zhou et al., 2021, 2022), REOF–sparse-GP flood surrogates that motivate localized EOF structure inside an LF–EOF–SGP pipeline (Wang et al., 2025), and regionalized LSG training for local velocity dimensionality-reduction error (Tan et al., 2025). Those precedents rule out “first zonal LSG” and any broad claim that regionalizing EOF/LSG is itself novel. Against these precedents, the unresolved question examined here is not whether LSG can be regionalized, but whether zoning changes LSG-Max depth skill when total retained-mode capacity is matched, whether any gain can be localized within the representation–mapping pipeline, and where that benefit fails to generalize.

We therefore ask three methods questions under an audited retained-mode-budget protocol, area-weighted metrics, and train-only partitioning: (RQ1) when does zoning help; (RQ2) why does it help when it does (representation, mapping, or their coupling); and (RQ3) when not, and can a simple training-data diagnostic identify those cases? We implement hydrodynamically zoned LSG-Max (maximum inundation surfaces) with rule and KMeans partitions on Carlisle, Chowilla, and Burnett River, and add residual-organization and stage-swap diagnostics. The contribution is therefore a capacity-controlled and falsifiable assessment of conditional zoning, rather than a new regionalization concept or a universal zoning prescription.

2. Methods

2.1 Cases and data

We use the public Fraehr multi-fidelity inundation benchmark (Figshare 24312658). Carlisle (UK; HF LISFLOOD-FP, LF HEC-RAS 2D) is the primary statistical case (9 events). Chowilla (Australia; MIKE 21) is treated as a boundary case. Burnett River (Australia; HF TUFLOW, LF HEC-RAS 2D) provides a 30-event leave-one-out contrast. Cell counts follow Fraehr geometry (Carlisle HF/LF $\approx$ 581,061 / 5,681; Chowilla $\approx$ 109,914 / 1,434; Burnett $\approx$ 780,785 / 15,256). Synthetic 30 $\times$ 40 Track A trees are excluded from all claims.

Table 1. Case roles, geometry, and paper role (Fraehr Figshare 24312658). Max-surface EOI (Carlisle / Chowilla / Burnett) = 0.057 / 0.116 / 0.957

Case	Country	Hydro model (HF)	Events used / available	HF / LF cells	Paper role
Carlisle	UK	LISFLOOD-FP	9 / 9	581,061 / 5,681	Primary; 9-fold LOOCV
Chowilla	Australia	MIKE 21	12 / 31	109,914 / 1,434	Boundary: LSG degrades vs LF-only
BurnettRV	Australia	TUFLOW	12 (three-case split); 30 (LOOCV) / 74 available	780,785 / 15,256	Contrast; Rule not favoured on LOOCV

2.2 Global and zoned LSG-Max

Wet cells use depth threshold 0.03 m. Global LSG-Max fits EOF on the wet HF training stack, projects LF onto the same modes, and maps LF $\rightarrow$ HF coefficients with sklearn Gaussian Process Regression (GPR). Zoned LSG-Max first partitions wet cells (rule: training max depth, inundation frequency, and an optional training LF–HF mean-absolute-residual hotspot override; or KMeans K = 4), then runs per-zone EOF+GP under a shared total mode budget B. Zoning features, EOF bases, and GPs are fit on training events only.

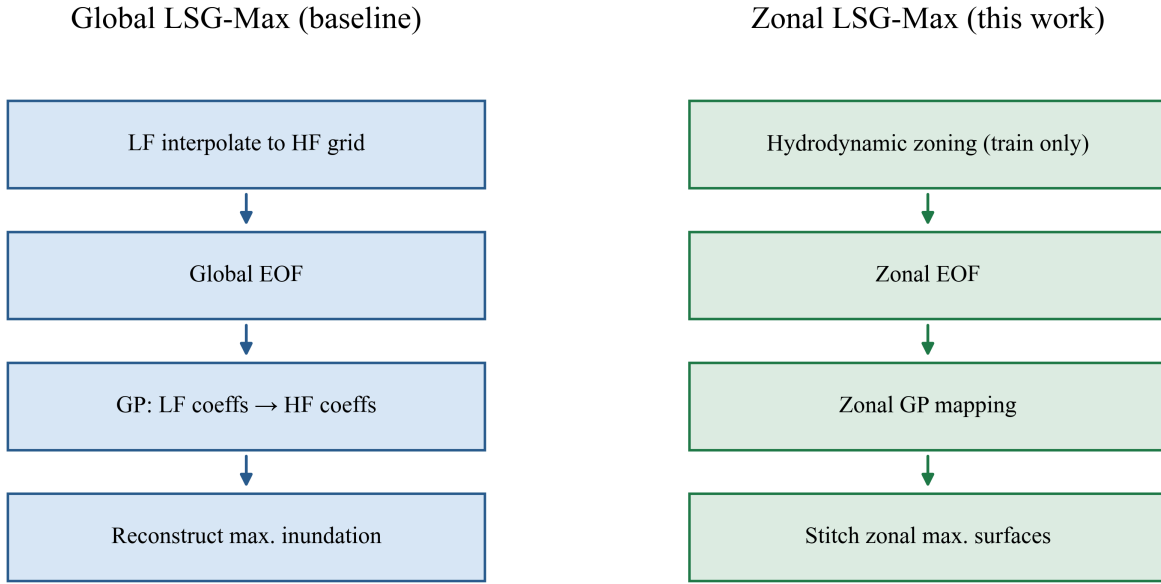


Figure 1. Hydrodynamically zoned LSG-Max workflow under equal mode budget.

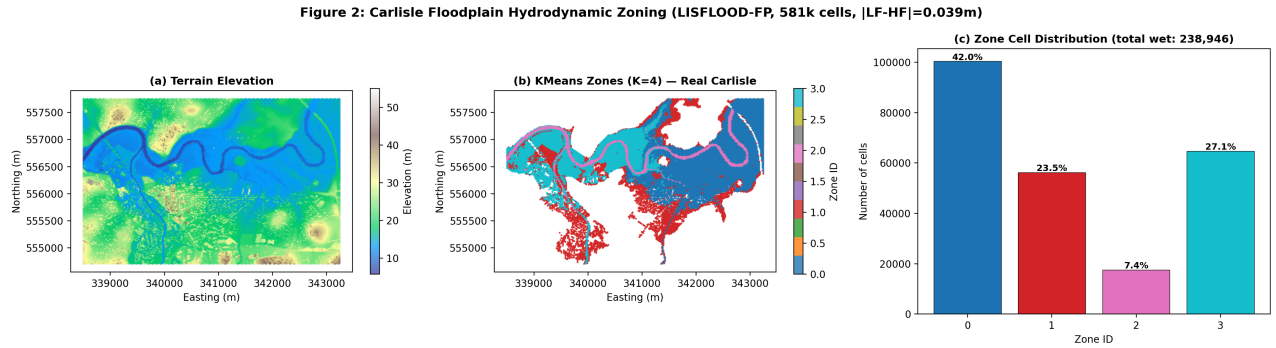


Figure 2. Rule and KMeans zone maps on the Carlisle wet floodplain.

2.3 Fair comparison protocol

Audited retained-mode-budget protocol (Carlisle): matched equal total retained-mode budgets at $B \in \{4, 6\}$; Global $B = 8$ is retained only as a budget-audit exception (requested 8, realized 7 = MISMATCH), not as a strict equal- B claim. Matched budgets mean equal retained EOF capacity, not equal wall-clock cost or equal GP complexity. Area-weighted RMSE, MAE, bias, and CSI use geometric cell areas. All fitted quantities are derived from training events only. The archived CLEAN_PASS leakage audit currently documents Carlisle Fold 1 (official split disjointness plus train-only zoning/EOF/GP checklist); automated all-fold and multi-case leakage audits remain 【待补充】.

2.4 Residual organization and stage-swap

First-order EOI is the variance of zone-mean training $|LF-HF|$ residuals divided by the variance of cellwise residuals on the training wet mask. We treat the EOI-as-switch test as an exploratory falsification of a plausible first-order training-data diagnostic, not as prospective selector validation. Second-order ZGG compares local-mode versus restricted-global-mode variance explanation at equal local rank, and an equal-budget pure-EOF reconstruction oracle tests whether zoning improves HF reconstruction without GP learning (current oracle RMSE is cell-unweighted; area-weighted oracle sensitivity remains 【待补充】). Stage-swap arms (GG/ZZ/GZ/ZG) cross global versus zonal EOF coordinates with global versus zonal GP stacks; they are not

four competing production models. GZ restricts and QR-orthonormalizes leading global modes within each zone before zonal GP mapping; ZG concatenates zonal expansion coefficients into a shared global-style GP stack. Both GZ and ZG are diagnostic approximations rather than algebraically exact stage substitutions; accordingly, the stage-swap is used to reject single-stage explanations, not to estimate a unique causal contribution of each stage.

2.5 Statistical claims

Principal Carlisle paired stability analysis: 9-fold event LOOCV at $B = 4$ (and secondary $B = 6$). Official two-fold bootstrap is reported as a benchmark-compatible sensitivity check and is not promoted as the primary claim. Burnett uses 30-fold event LOOCV at $B = 4$. Uncertainty uses bootstrap confidence intervals on $\Delta\text{RMSE} = \text{RMSE}_{\text{global}} - \text{RMSE}_{\text{zonal}}$.

3. Results

3.1 Equal-budget Carlisle curves

At $B = 4$, Global / Rule / KMeans area-weighted RMSE = 0.1464 / 0.0964 / 0.1015 m (Rule -34.2% vs Global). Across the audited budget sweep, Global RMSE increased from 0.1464 to 0.2588 and 0.3527 m, while Rule increased from 0.0964 to 0.1256 and 0.1790 m. $B = 4$ and 6 are strict matched-budget comparisons; the nominal Global $B = 8$ point realized seven modes and is retained only as a budget-audit exception. We report the Global– B rise as an observation under this audited protocol; component-level attribution (for example, event-noise fitting) is not uniquely identified here. CSI does not show a matching zonal win: LF-only CSI (0.9145) exceeds LSG CSI at $B = 4$, so zoning improves depth RMSE more than wet–dry extent.

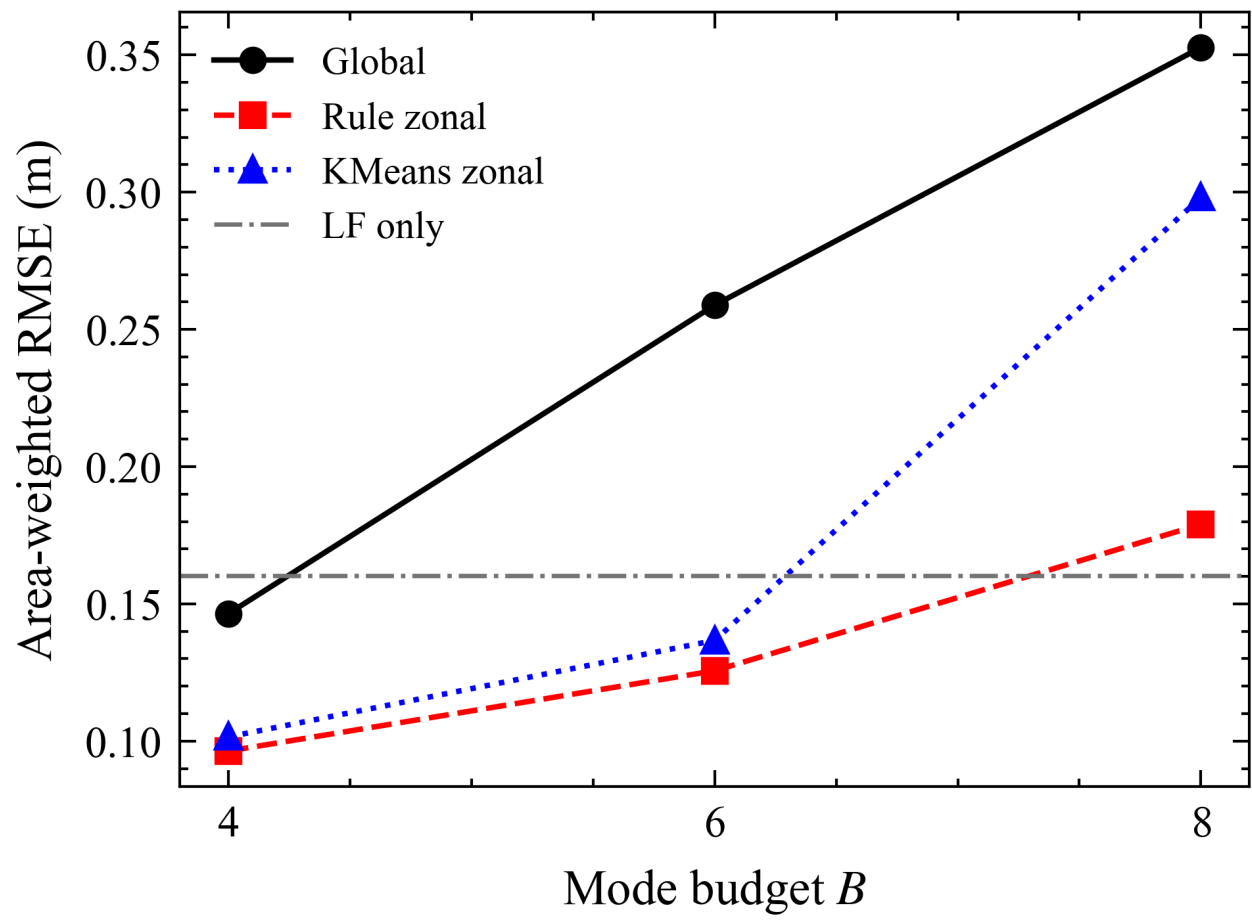


Figure 3. Carlisle area-weighted depth RMSE versus total mode budget B .

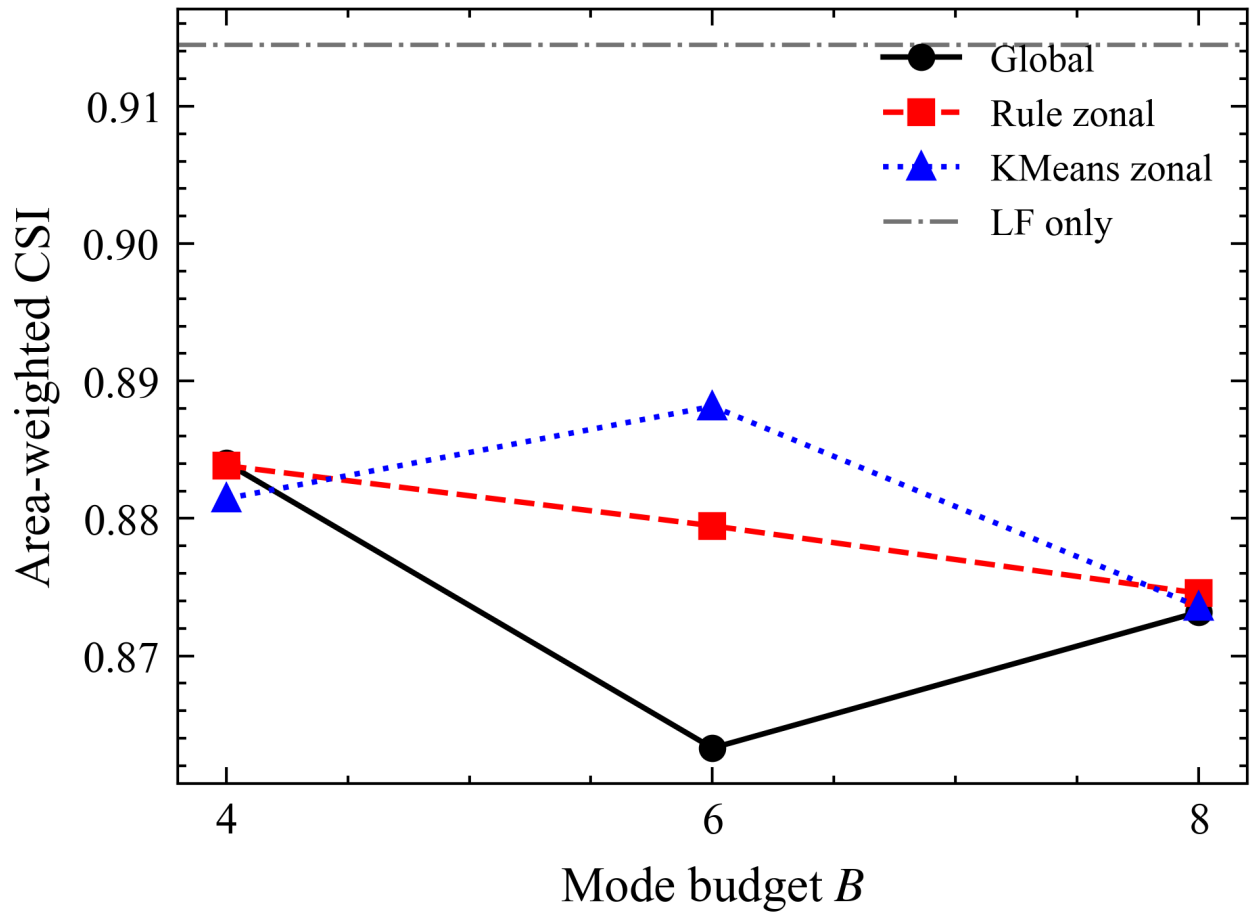


Figure 9. Carlisle area-weighted CSI versus mode budget B .

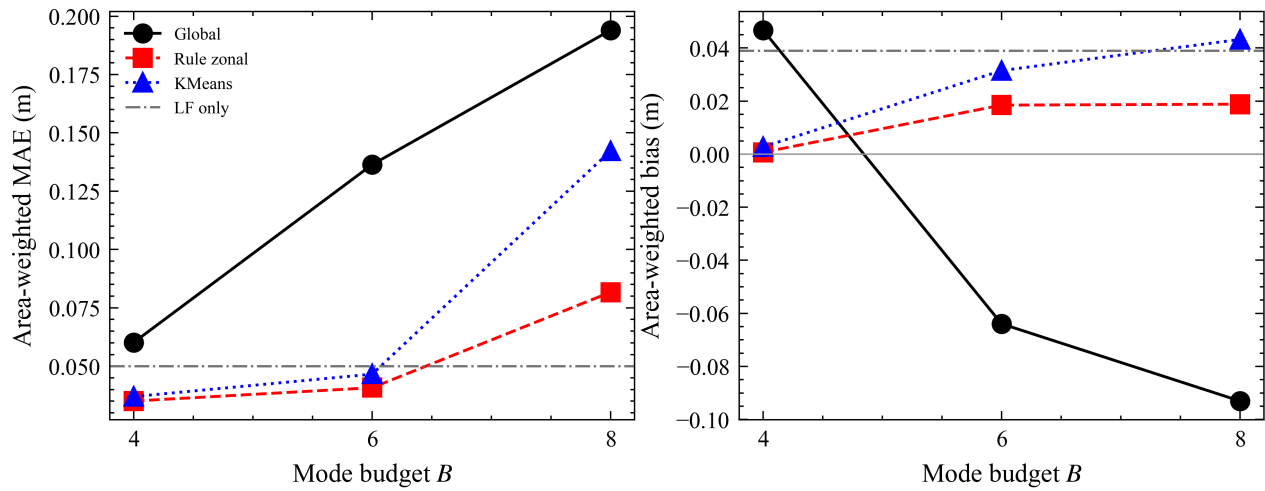


Figure 13. Carlisle MAE and bias versus mode budget.

3.2 Event-level statistics

Carlisle $B = 4$ LOOCV: 9/9 folds improved; mean $\Delta\text{RMSE} = 0.0821$ m; 95% CI [0.0155, 0.1987]. $B = 6$: 7/9; mean 0.0606 m; CI [0.0032, 0.1618]. Official two-fold: mean $\Delta\text{RMSE} = 0.0045$ m; CI [-0.0073, 0.0134]; significant = false. Burnett $B = 4$ 30-fold: mean Global 1.7479 m vs Rule 1.8260 m; $\Delta\text{RMSE} = -0.0781$ m; 6/30; significant = false.

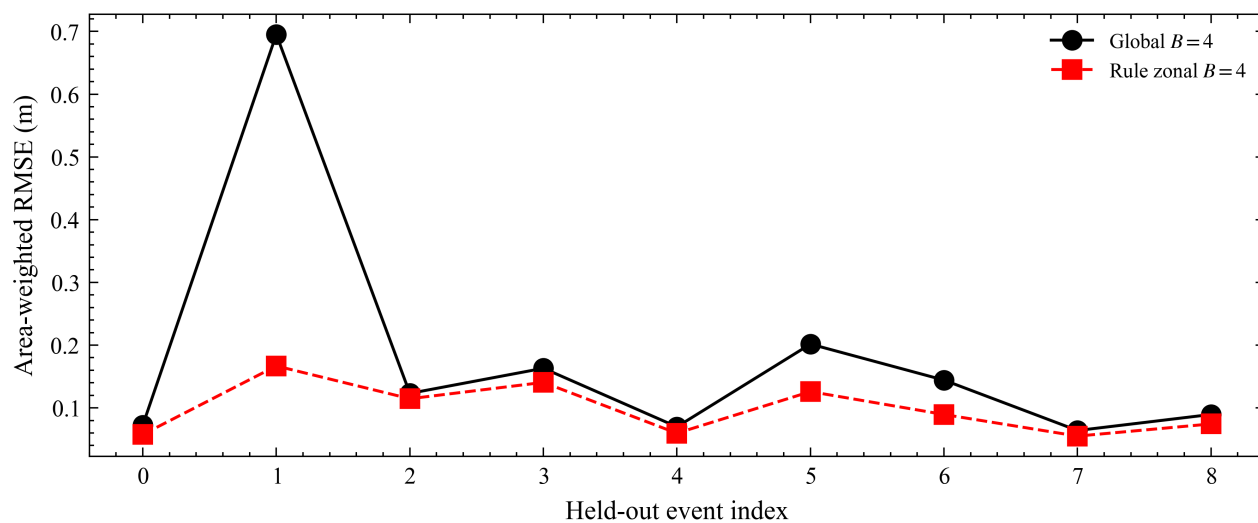


Figure 8. Carlisle per-event Δ RMSE with bootstrap uncertainty.

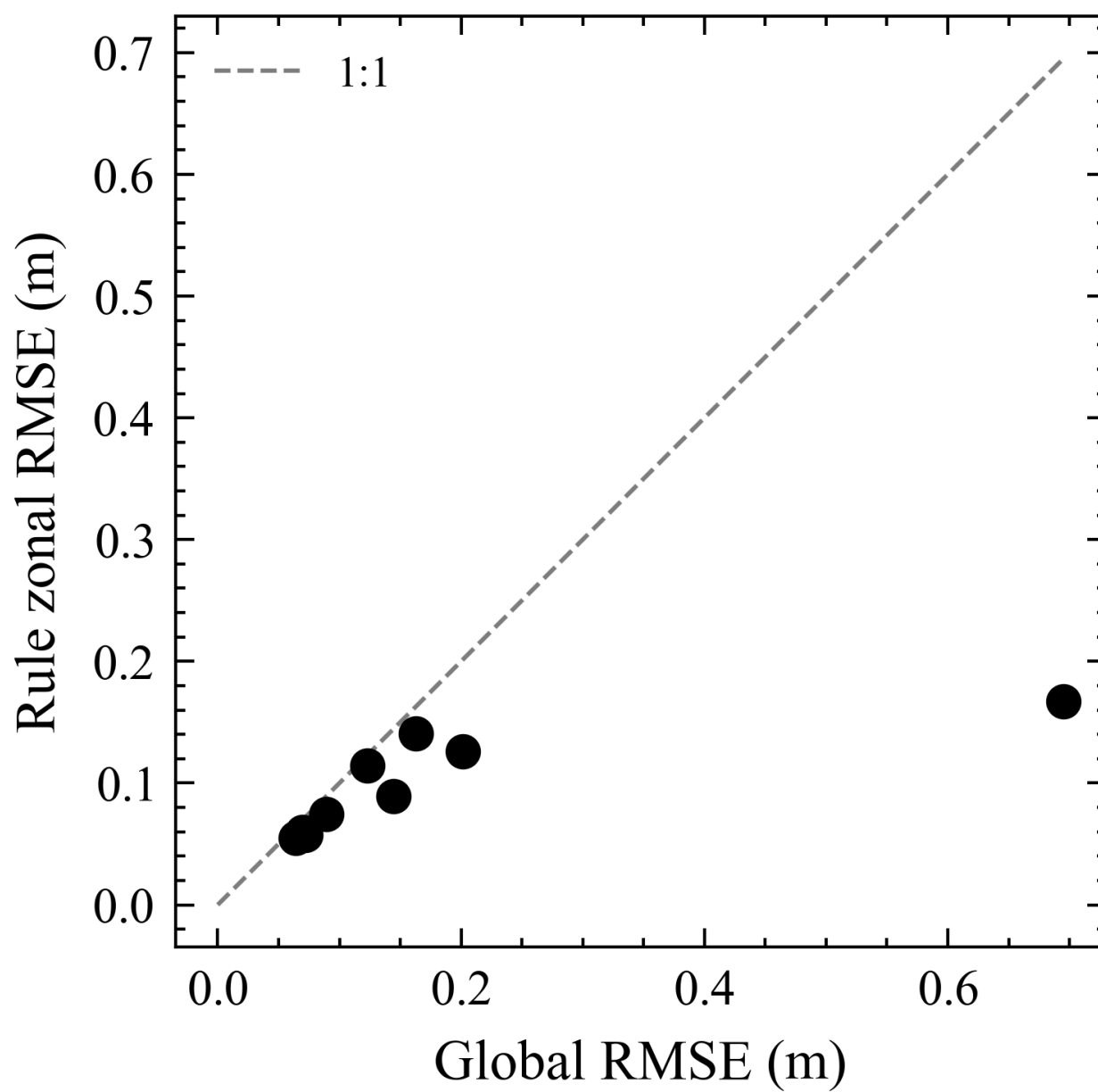


Figure 11. Carlisle LOOCV Global versus Rule RMSE scatter.

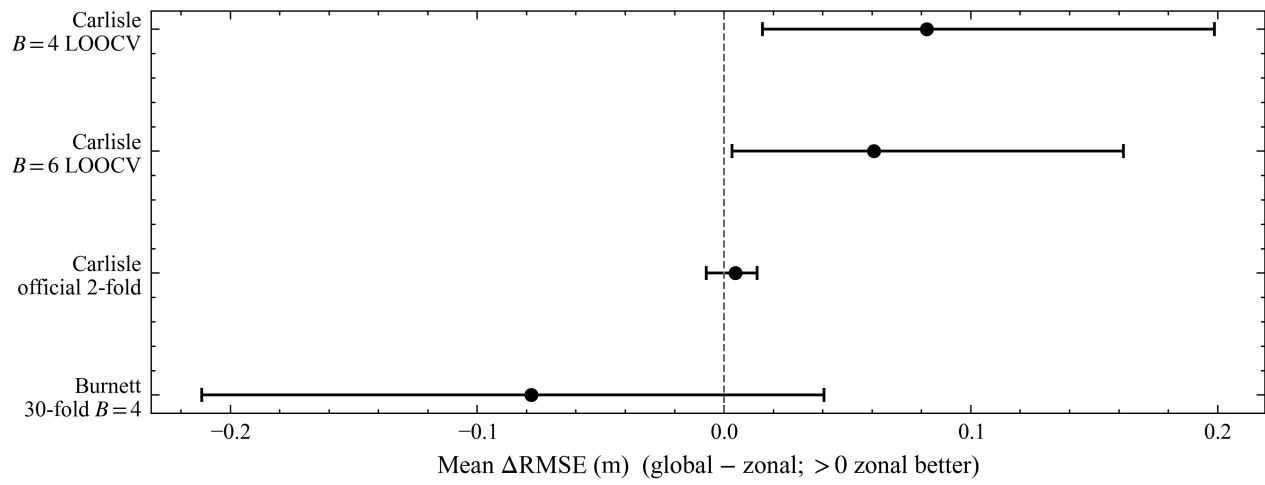


Figure 12. Statistical confidence intervals for ΔRMSE contrasts.

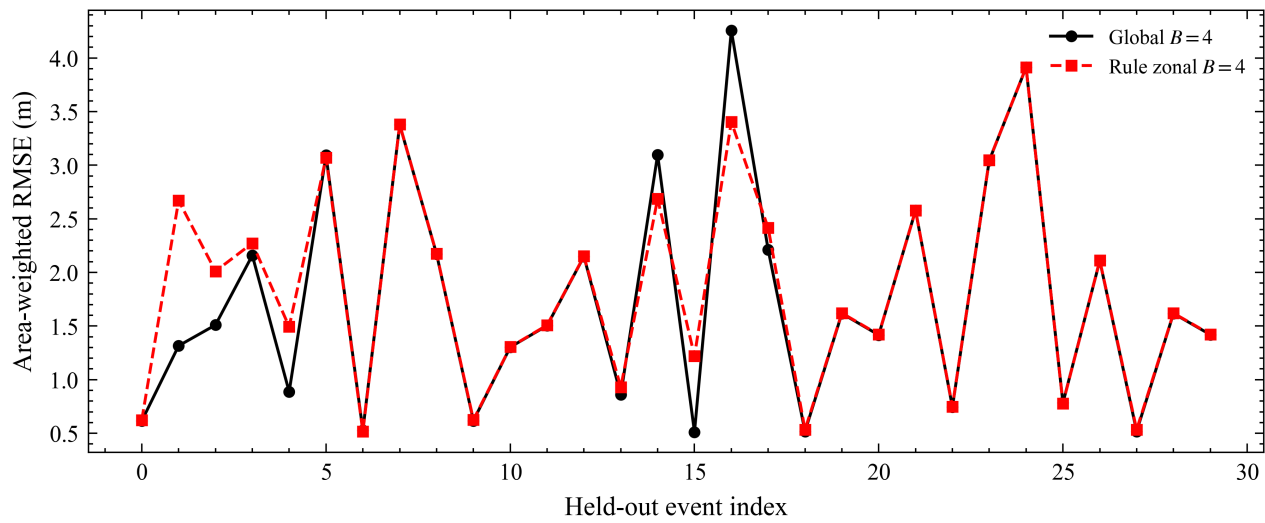


Figure 10. Burnett River 30-fold event LOOCV summary.

3.3 Three-case pattern

Table 2. Three-case area-weighted depth RMSE (m) at $B = 4$

Case	LF-only	Global $B=4$	Rule $B=4$	Pattern
Carlisle	0.1602	0.1464	0.0964	Rule $<$ Global $<$ LF (RMSE; lower better)
Chowilla	0.3926	2.5606	2.5614	LF best; LSG RMSE worse than LF-only (upstream boundary)
Burnett (12-event fixed split)	2.2323	1.6120	1.6122	Global $\approx$ Rule (not the 30-fold LOOCV protocol)

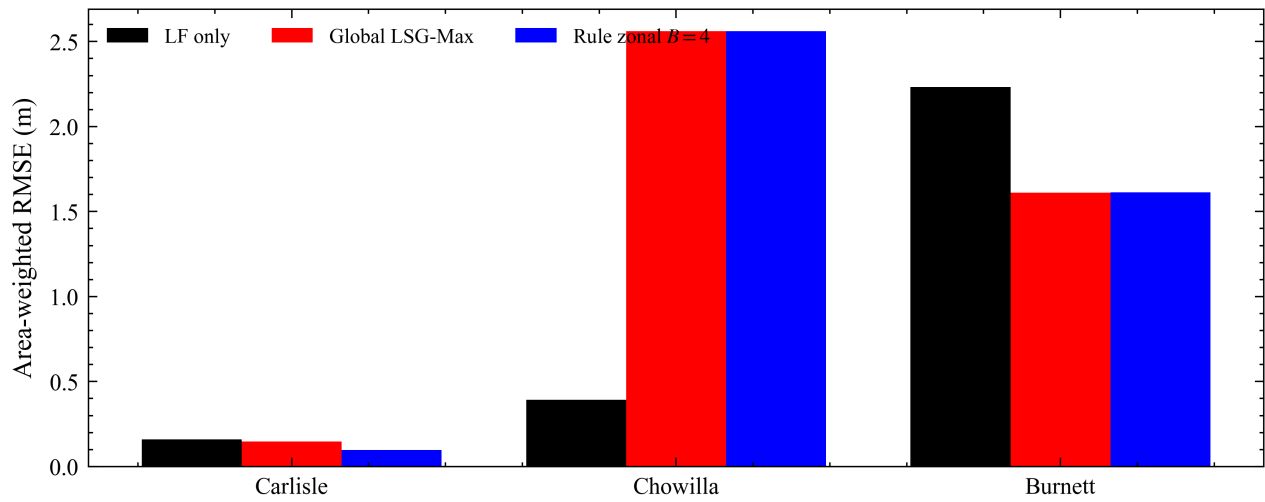


Figure 4. Three-case RMSE comparison (Carlisle, Chowilla, Burnett).

3.4 EOI, second-order, and stage-swap

Max-surface EOI: Carlisle 0.057, Chowilla 0.116, Burnett 0.957. Zoning benefit does not increase with EOI. Modal EOI registry shows $ZGG > 0$ with equal-budget pure-EOF oracle $\Delta RMSE < 0$ on all three cases ($ZGG_POSITIVE_ORACLE_LOSS$), ruling out “zoning helps merely by truncating HF EOF better.” Stage-swap LOOCV means: $GG \approx 0.180$, $ZZ \approx 0.098$, $GZ \approx 0.098$, $ZG \approx 0.101$ m. These results reject a pure-EOF-truncation explanation and are consistent with zonal structure acting through the coupled representation–mapping pipeline; they do not uniquely attribute the gain to one localized stage or identify GP locality alone as the mechanism.

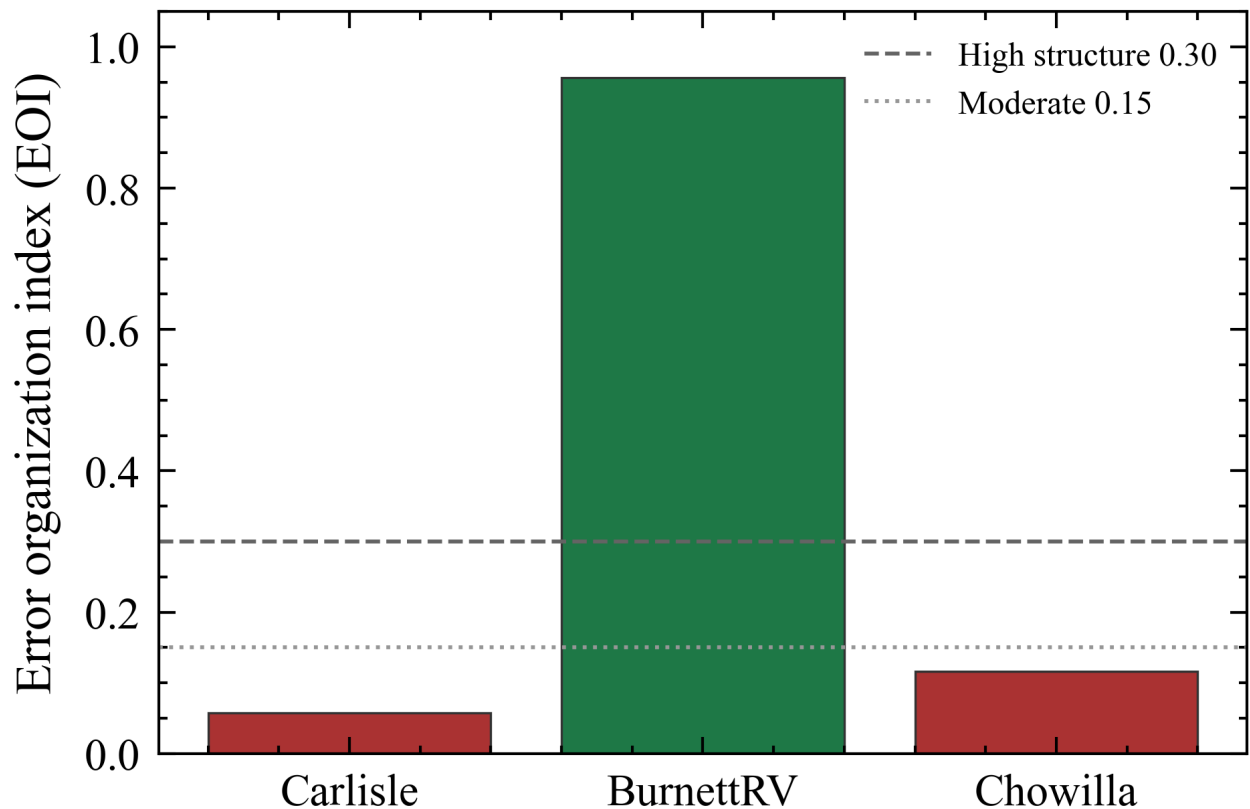


Figure 14. Max-surface residual organization index (EOI) by case.

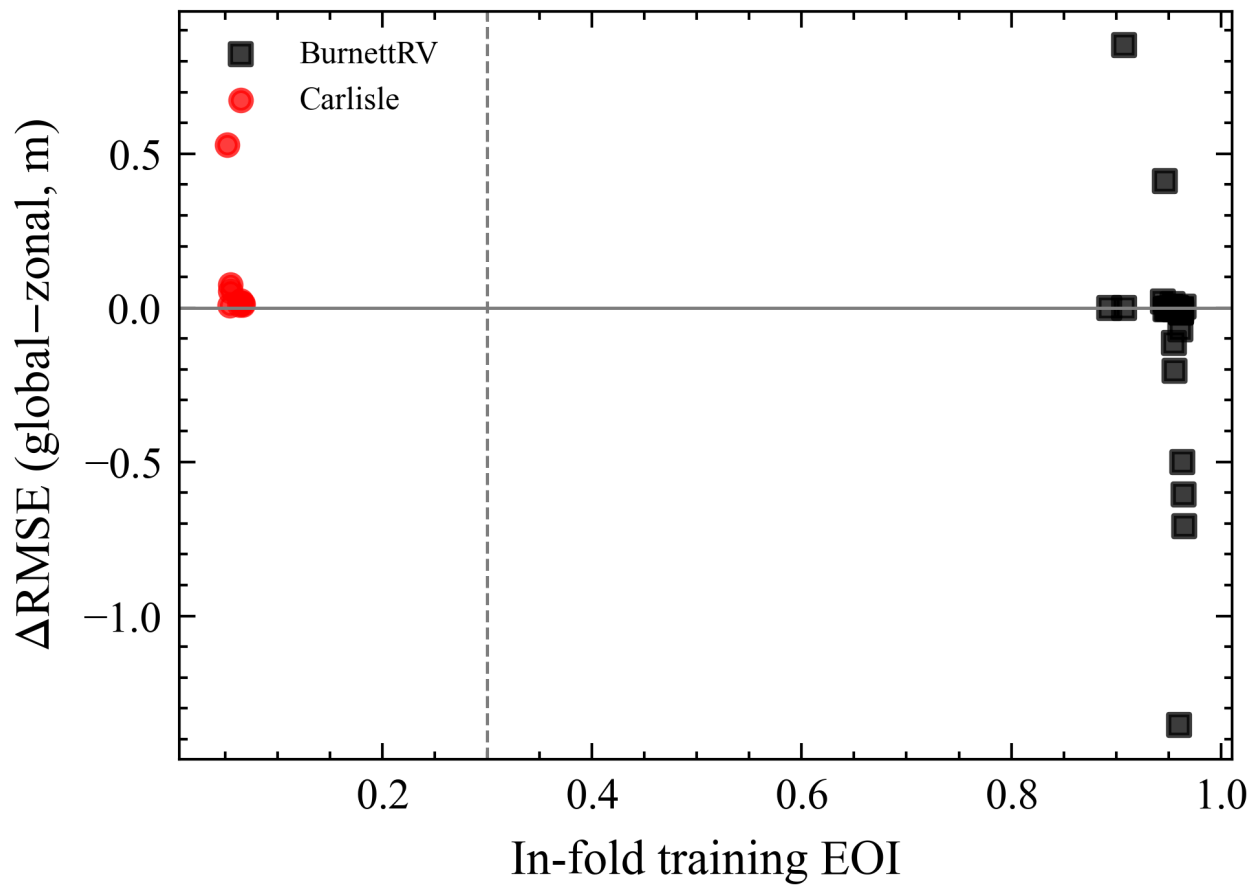


Figure 15. EOI versus zoning ΔRMSE (does not rank zoning gains).

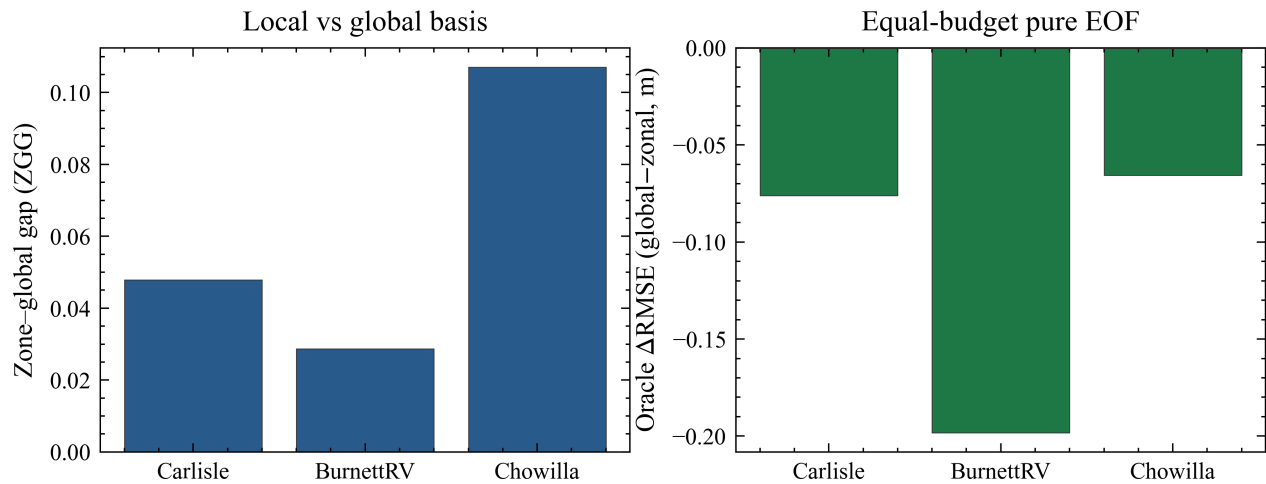


Figure 19. Second-order modal EOI / pure-EOF oracle diagnostics.

3.5 Published MaxWD contrast and robustness probes

This comparison is used only as a protocol-level sanity check and is not a head-to-head comparison with a locally reproduced LSG-TS implementation. On the official nine-fold MaxWD R^2 protocol, rule LSG-Max reaches 0.988 versus published LSG-TS 0.990 (Global Max $\approx$ 0.915). LF coarsening and channel-distance zoning probes (Figs. 17–18) support robustness narratives but are secondary to the equal-B LOOCV claim.

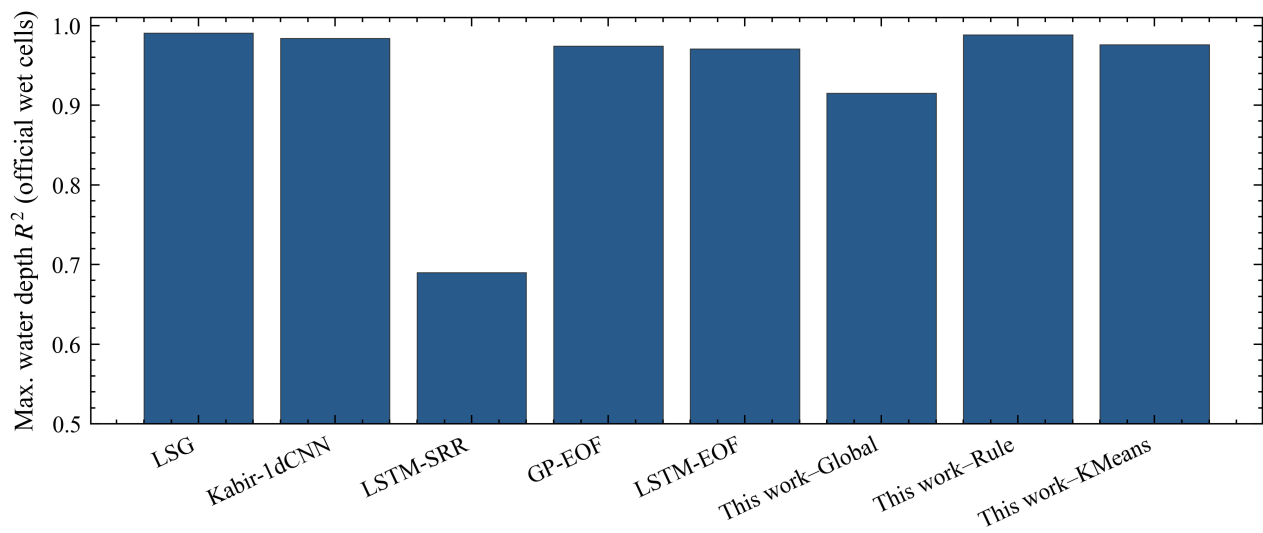


Figure 16. Official nine-fold MaxWD R^2 protocol contrast.

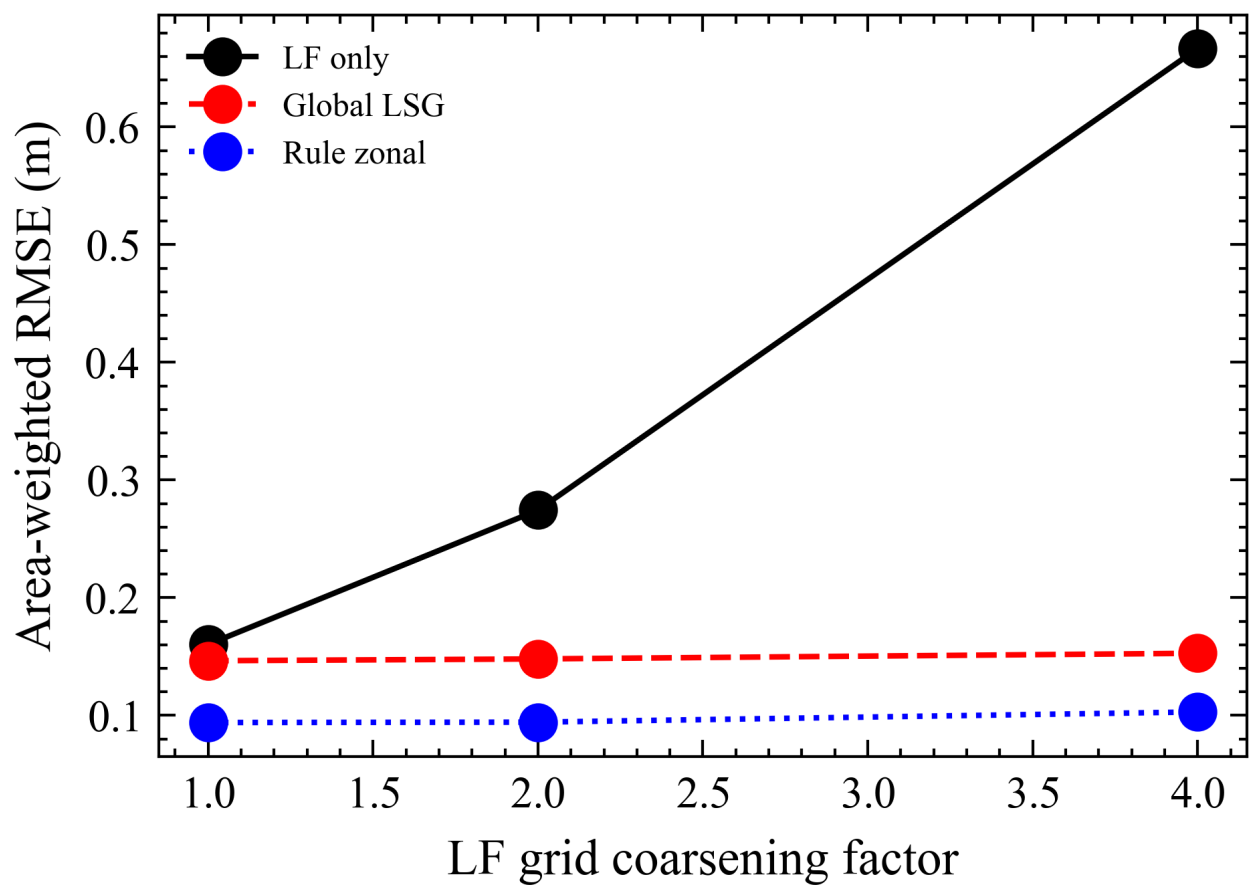


Figure 17. LF coarsening robustness probe.

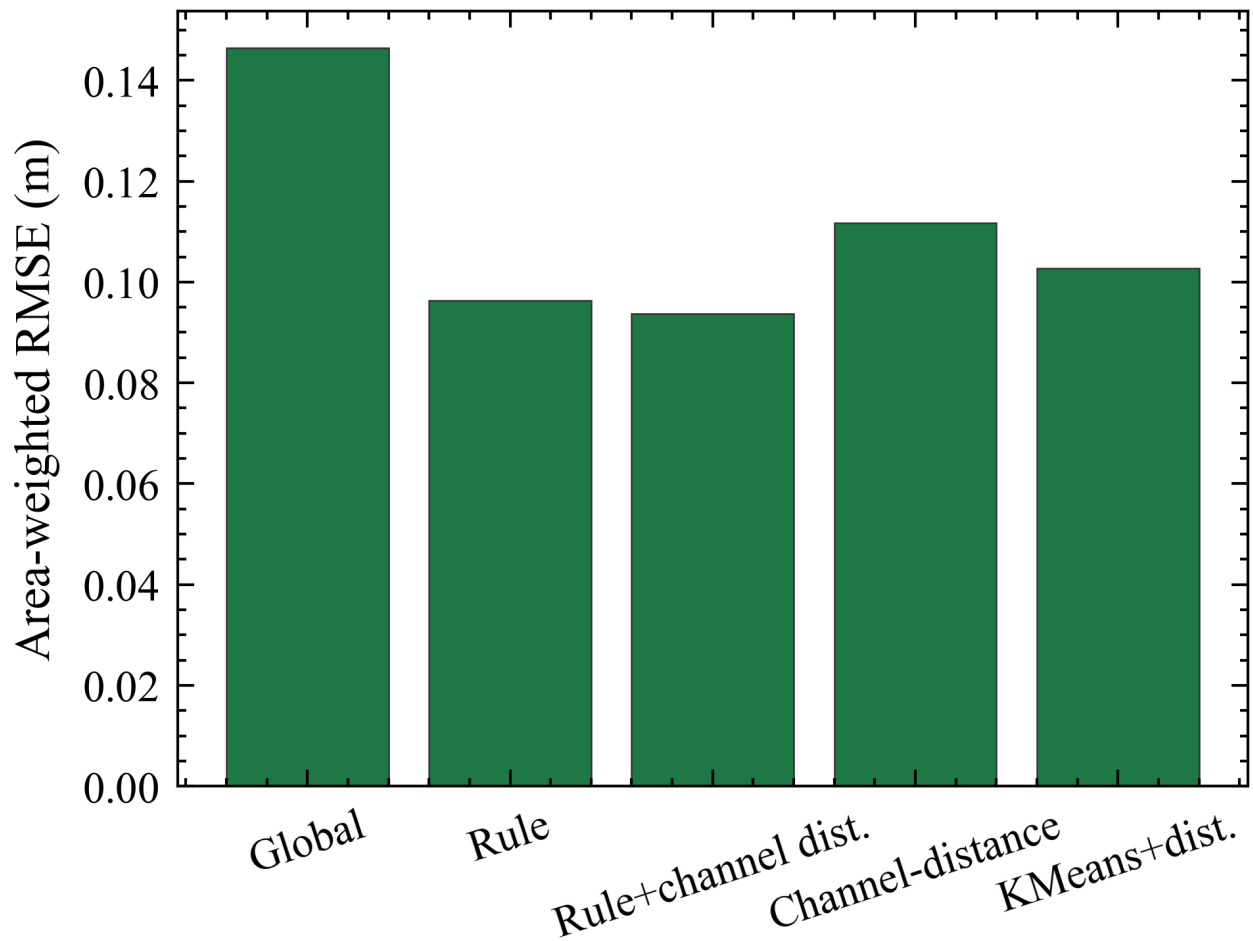


Figure 18. Channel-distance zoning robustness probe.

4. Discussion

Thesis. Spatial zoning is not universally advantageous, but a global EOF representation can be performance-sensitive under constrained retained-mode capacity. Under an audited retained-mode-budget protocol, the value of zoning is conditional on how spatial structure interacts with the coupled reduced-representation and LF-to-HF mapping pipeline; simple field heterogeneity alone is insufficient to decide when zoning should be used.

Q1. When does zoning help under equal representation capacity?

At Carlisle under equal $B = 4$, rule zoning reduced area-weighted depth RMSE with fold-consistent improvement (9/9 LOOCV). The gain concerns depth-error organization under limited capacity, not inventing wet cells that the LF already captures well (CSI remains LF-led). The official two-fold contrast was non-significant and is kept as a sensitivity check. Burnett 30-fold leave-one-out remains the equal-budget non-benefit contrast. This is the conditional claim, not evidence that zoning always wins.

Q2. Why can zoning help when it does?

The diagnostics progressively narrow, but do not uniquely identify, the mechanism. First-order EOI does not explain the cross-case pattern, and the pure-EOF oracle argues against a truncation-only account. The approximate Carlisle stage-swap is instead consistent with zonal structure entering the coupled reduced-

representation and LF-to-HF mapping problem, while providing no basis for assigning the gain uniquely to EOF localization or GP localization. The non-monotonic Global-B behaviour is reported as a related capacity observation, not as the Discussion spine.

Q3. When should zoning not be used?

The present results do not support zoning from residual heterogeneity alone. Across the three tested cases, max-surface EOI provides no reliable threshold for choosing between global and zonal LSG: Burnett combines the highest EOI with no Rule advantage, whereas Carlisle shows the clearest benefit at the lowest EOI. Chowilla defines a different boundary, because LSG itself performs substantially worse than LF-only; in such a case, whether statistical LF-to-HF correction is appropriate precedes the question of how its reduced representation should be partitioned. EOI has been ruled out as a sufficient first-order training-data diagnostic, but the present experiments do not yet establish a transferable criterion for choosing between global and zonal LSG.

Rival explanations rejected or weakened: (i) accidental unequal budgets (audited); (ii) test leakage into zoning (train-only fits; archived Carlisle Fold 1 CLEAN_PASS); (iii) GP-only localization as the unique cause ($ZG \approx ZZ$ as well as $GZ \approx ZZ$); (iv) official two-fold as primary evidence (CI includes zero).

Limitations

Several limitations bound these conclusions. The experiments use LSG-Max with sklearn GPR rather than a canonical gpflow/SGPR or locally reproduced LSG-TS implementation, and the Brisbane case has not been evaluated; backend-equivalence sensitivity remains 【待补充】. Carlisle is the only clear zoning-positive public case, so no transferable operational selector is claimed. Chowilla archive checksum/datum provenance remains unresolved, and Burnett KMeans LOOCV and the full 74-event evaluation remain incomplete. Historical temporal EOI (≈ 0.51) uses a different protocol and is SI-only. In addition, the archived CLEAN_PASS audit currently documents Carlisle Fold 1 with code-reviewed train-only checks rather than an automated audit over every fold and case; broader audit coverage remains 【待补充】.

5. Conclusions

1. Under matched retained-mode capacity, Carlisle shows that a global EOF representation can be performance-sensitive: Rule zoning reduces area-weighted RMSE from 0.1464 to 0.0964 m at $B = 4$ and improves all nine leave-one-out folds.
 2. In the Carlisle budget sweep, increasing the requested global mode count did not recover the $B = 4$ zonal advantage; the nominal $B = 8$ point is retained as an audited mismatch (Global realized 7 modes) rather than a strict equal-B comparison.
 3. Zoning benefit is conditional: Burnett 30-fold LOOCV does not favour Rule; Chowilla is an upstream LSG-vs-LF applicability boundary where LSG RMSE exceeds LF-only.
 4. Across these cases, first-order max-surface EOI does not select zoning; mechanism evidence rules out pure-EOF truncation and does not uniquely pin the gain to one stage (Carlisle stage-swap).
 5. Official two-fold non-significance must be stated; it is a sensitivity check, not the principal paired claim.
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Data and code availability (draft)

- Data: Fraehr et al. public benchmark, Figshare 24312658.
 - Code and artefacts: github.com/Coucou2016/202606-JOH-zonal-LSG (public; raw hydrodynamic HDF/NPZ excluded).
 - Numbers: `outputs/registry/result_manifest_v4.csv` and evaluation JSON under `outputs/evaluation/`.
 - Report generators: `scripts/95_final_submission_report.py`, `scripts/96_research_report_zh.py`, `scripts/99_full_report_zh.py`.
 - Figures: `outputs/figures/fig*.png` (SciencePlots + Times New Roman for Track B curves).
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References (independently verified seed list)

1. Fraehr, N., Wang, Q. J., Wu, W., & Nathan, R. (2022). Upskilling low-fidelity hydrodynamic models of flood inundation through spatial analysis and Gaussian Process learning. *Water Resources Research*, 58, e2022WR032248. doi:10.1029/2022WR032248
2. Fraehr, N., Wang, Q. J., Wu, W., & Nathan, R. (2023). Development of a fast and accurate hybrid model for floodplain inundation simulations. *Water Resources Research*, 59, e2022WR033836. doi:10.1029/2022WR033836
3. Fraehr, N., et al. (2024). Assessment of surrogate models for flood inundation: The physics-guided LSG model vs. state-of-the-art machine learning models. *Water Research*, 252, 121202. doi:10.1016/j.watres.2024.121202
4. Bentivoglio, R., Isufi, E., Jonkman, S. N., & Taormina, R. (2022). Deep learning methods for flood mapping: a review of existing applications and future research directions. *Hydrology and Earth System Sciences*, 26, 4345–4378. doi:10.5194/hess-26-4345-2022
5. Teng, J., Jakeman, A. J., Vaze, J., Croke, B. F. W., Dutta, D., & Kim, S. (2017). Flood inundation modelling: A review of methods, recent advances and uncertainty analysis. *Environmental Modelling & Software*, 90, 201–216. doi:10.1016/j.envsoft.2017.01.006
6. Bates, P. D. (2022). Flood inundation prediction. *Annual Review of Fluid Mechanics*, 54, 287–315. doi:10.1146/annurev-fluid-030121-113138
7. Lu, J., Wang, Q. J., Fraehr, N., Xiang, X., & Wu, X. (2025). Choice of Gaussian Process kernels used in LSG models for flood inundation predictions. *Journal of Hydrology*, 655, 132949. doi:10.1016/j.jhydrol.2025.132949
8. Carreau, J., & Guinot, V. (2021). A PCA spatial pattern based artificial neural network downscaling model for urban flood hazard assessment. *Advances in Water Resources*, 147, 103821. doi:10.1016/j.advwatres.2020.103821
9. Zhou, Y., Wu, W., Nathan, R., & Wang, Q. J. (2021). A rapid flood inundation modelling framework using deep learning with spatial reduction and reconstruction. *Environmental Modelling & Software*, 143, 105112. doi:10.1016/j.envsoft.2021.105112
10. Zhou, Y., Wu, W., Nathan, R., & Wang, Q. J. (2022). Deep learning-based rapid flood inundation modeling for flat floodplains with complex flow paths. *Water Resources Research*, 58, e2022WR033214. doi:10.1029/2022WR033214
11. Tan, Z., Xu, D., Taraphdar, S., Ma, J., Bisht, G., & Leung, L. R. (2025). An efficient hybrid downscaling framework to estimate high-resolution river hydrodynamics. *Hydrology and Earth System Sciences*, 29, 3833–3852. doi:10.5194/hess-29-3833-2025

12. Wang, R., Lian, J., Yuan, X., Tian, F., Li, K., & Liu, Z. (2025). Rapid simulation of floods by considering the spatial and temporal characteristics of inundation. *International Journal of Disaster Risk Science*, 16, 481–495. doi:10.1007/s13753-025-00642-5
13. Fraehr, N., Wang, Q. J., Wu, W., & Nathan, R. (2023). Supercharging hydrodynamic inundation models for instant flood insight. *Nature Water*, 1, 835–843. doi:10.1038/s44221-023-00132-2
14. Wang, W., Wang, Q. J., & Nathan, R. (2026). Strategies for predicting flood inundation in a large and complex floodplain based on low-fidelity hydrodynamic models. *Water Resources Research*. doi:10.1029/2025WR042481

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Assumptions / missing inputs

- GP backend remains sklearn until gpflow/SGPR is installed and re-run.
- Figure binaries may still be regenerating; filenames above are the contract with the figure workline.
- Author list, funding, and exact target journal wording TBD.